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Letter

Machine Learning Accelerated Design of Self-Assembled Monolayers for High-Performance Perovskite Solar Cells

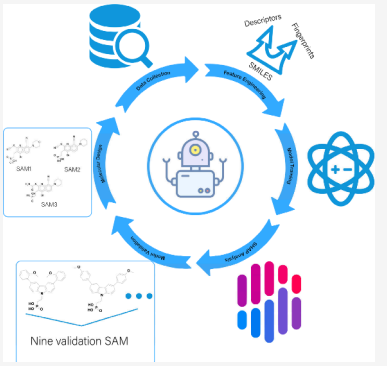
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ABSTRACT: Self-assembled monolayers (SAMs) have emerged as a new generation of hole transport materials (HTMs) for perovskite solar cells (PSCs), particularly in inverted architectures. Compared to conventional HTMs, SAMs demonstrate superior power conversion efficiency (PCE) and enhanced operational stability. However, the current discovery of SAMs still relies heavily on empirical trial-and-error approaches, suffering from long development cycles, high costs, and low success rates. Here we present a novel machine learning (ML) platform for accelerated SAM discovery and design. We constructed a comprehensive feature space combining RDKit molecular descriptors and Morgan fingerprints, and then systematically evaluated various ML algorithms. Multiple evaluation metrics were used to assess model reliability. The results demonstrate that the RDKit-based XGBoost model achieved optimal performance with a root-mean-square error (RMSE) of 1.862, a coefficient of determination (R^2) of 0.5058, a Pearson correlation coefficient (r) of 0.8161 and a mean absolute error (MAE) of 1.528. Then, SHapley Additive exPlanations (SHAP) analysis further elucidated the structure−property relationships between key molecular features and photovoltaic performance. The SHAP values revealed that the top five most important features were all RDKit descriptors, specifically EState_VSA5, fr_benzene, EState_VSA2, SlogP_VSA1, and Chi0v. The external validation using recently reported SAM molecules demonstrated remarkable prediction accuracy. The relative errors between predicted and experimental PCE values were mostly within 10%, with the minimum being only 0.55%. Meanwhile, three new SAM molecules were designed based on the model, with the highest predicted PCE approaching 27%. Therefore, this work provides an efficient digital solution for SAM development, offering valuable guidance for accelerating the discovery of next-generation photovoltaic materials.



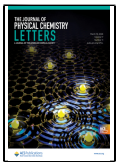
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Perovskite solar cells (PSCs) have attracted significant attention as a third-generation photovoltaic technology due to their high power conversion efficiency (PCE) and low manufacturing cost.[1−4] Recent advances have pushed the PCE of PSCs beyond 27%.[5,6] Among interfacial materials, self-assembled monolayers (SAMs) have shown great potential as hole transport layers, especially in inverted architectures, owing to their minimal parasitic absorption, low material usage, and simple processing, which support the fabrication of large-area and stable devices.[7] However, current SAM design still relies on conventional synthesis methods, such as reactions with conjugated aldehydes, pyridines, or amine derivatives.[6] These approaches are not only time- and resource-intensive but also offer no guarantee of achieving high PCE. Thus, there remains a challenge to rapidly design and optimize SAMs for efficient perovskite solar cells.

In PSCs, SAMs are ultrathin, ordered molecular layers that self-assemble on substrates.[6] A typical SAM comprises three parts: an anchoring group, a spacer, and a functional group.[7] The anchoring group bonds chemically to the substrate, enabling film formation.[8] The spacer connects the anchor to the functional group,[7] which interfaces with the perovskite, enhances charge extraction, and modulates the substrate work function via its molecular dipole.[9,10] Recently, many studies have exploited SAMs in PSCs. For example, Zhang et al.[11] used SAMs with different anchors and spacers to tune perovskite interface properties in mixed Sn−Pb devices. Li et al.[12] applied Me-4PACz to stabilize the perovskite structure and passivate interfaces. Zhou et al.[13] synthesized benzothiadiazole-based SAMs with varied anchoring groups for interface modification. Despite these advances, SAM preparation remains largely trial-and-error, with no assurance of high device performance. Therefore, a method to rapidly screen high-PCE SAMs is urgently needed.

In recent years, the integration of artificial intelligence (AI) and materials science has opened new avenues for accelerated material discovery. Machine learning (ML), a cornerstone of AI, has demonstrated remarkable capability in establishing quantitative structure−property relationships (QSPRs) and predicting key material characteristics.[14] Within perovskite

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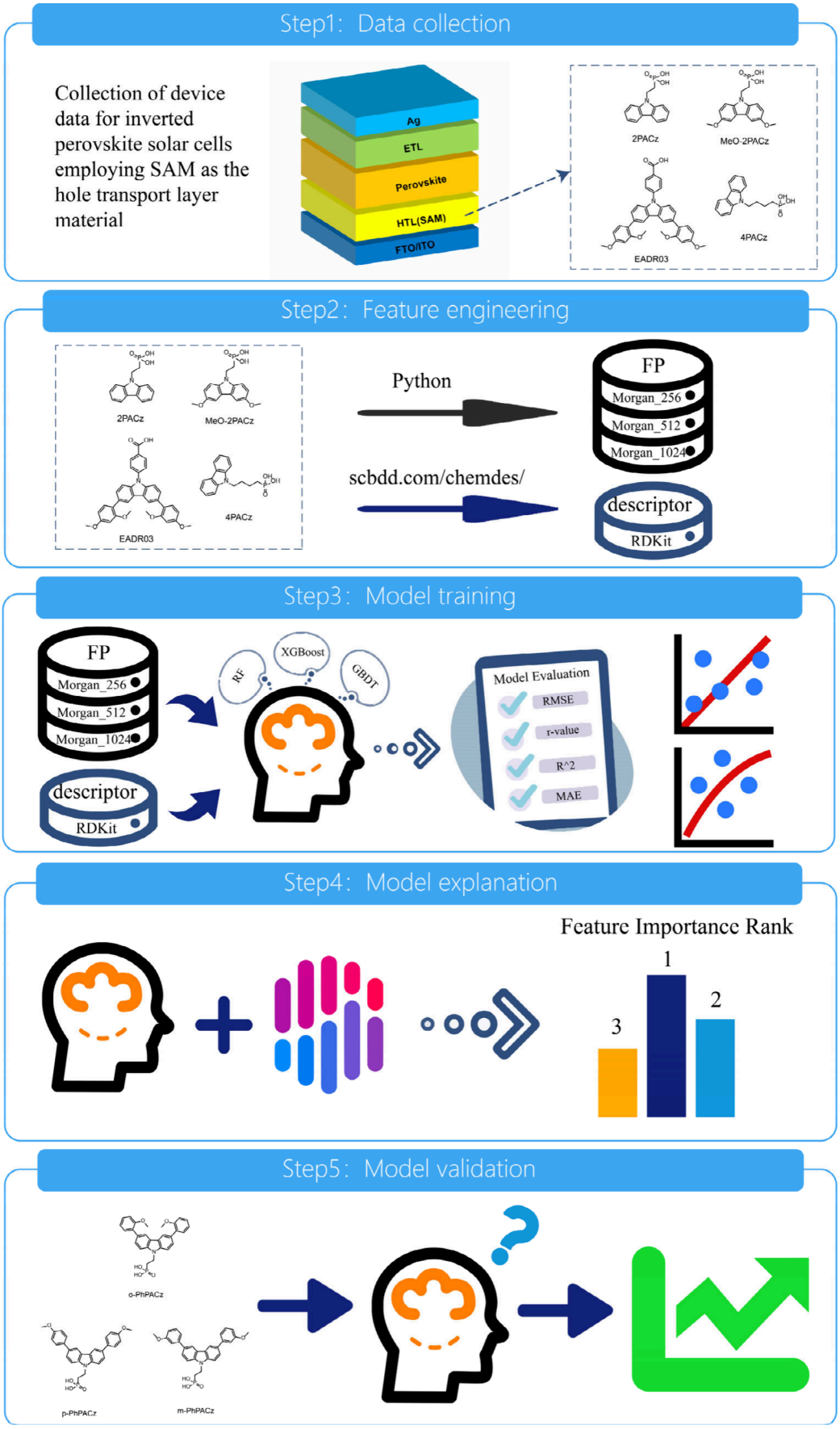


Figure 1. Workflow of this research work.

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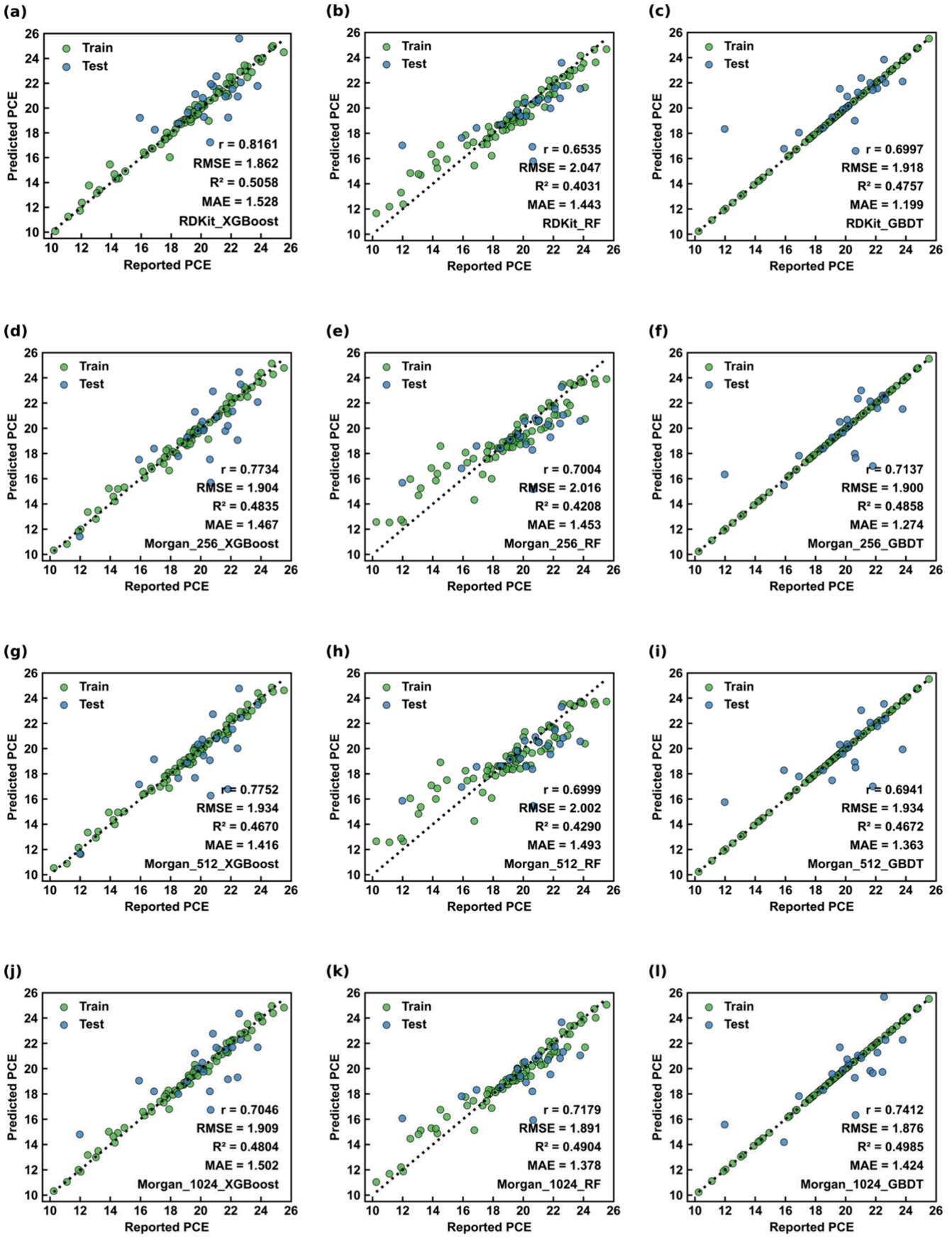


Figure 2. Prediction results of the 12 trained models. (a) RDKit_XGBoost. (b) RDKit_RF. (c) RDKit_GBDT. (d) Morgan_256_XGBoost. (e) Morgan_256_RF. (f) Morgan_256_GBDT. (g) Morgan_512_XGBoost. (h) Morgan_512_RF. (i) Morgan_512_GBDT. (j) Morgan_1024_XGBoost. (k) Morgan_1024_RF. (l) Morgan_1024_GBDT.

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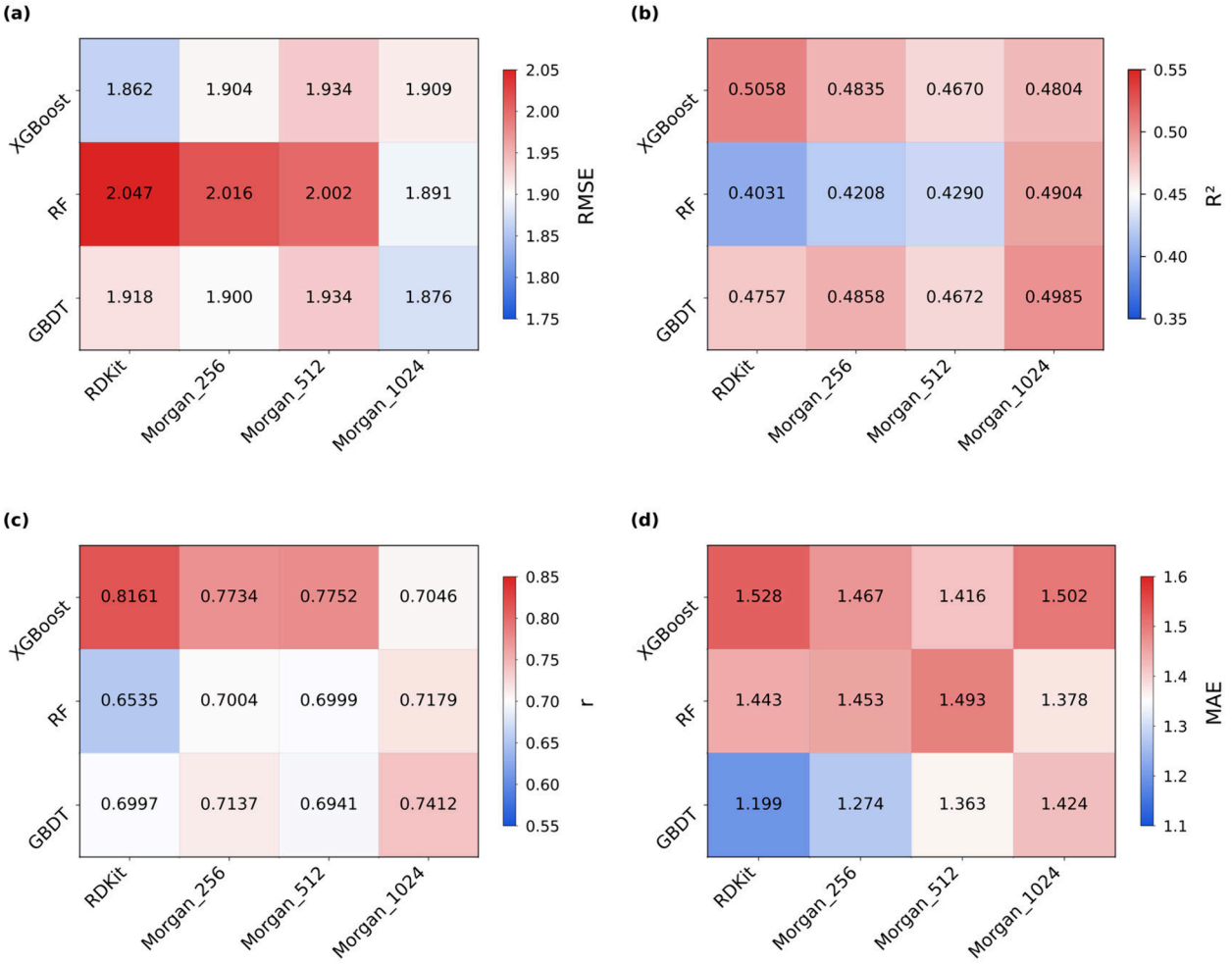


Figure 3. Performance metrics of all models. (a) The RMSE of all models. (b) The R^2 of all models. (c) The r of all models. (d) The MAE of all models.

research, ML has been successfully applied to predict properties such as bandgap,[15] conductivity,[16] and stability,[17] among others. Our research team has also contributed to related ML-driven studies on perovskites.[18−22] Beyond property prediction, ML aids in high-throughput fabrication optimization,[23] thin-film analysis,[24] and understanding the role of molecular additives.[25] These successes underscore ML's potential to transform the design cycle of functional materials like SAMs. By learning from existing experimental data, ML models can uncover hidden structure-performance correlations and rapidly screen vast chemical spaces, thereby drastically reducing the need for costly and time-consuming experimental iterations.

In this work, we present a data-driven ML framework specifically tailored for the accelerated discovery and design of high-performance SAMs for inverted PSCs. We constructed a curated data set of SAM-based devices and systematically evaluated multiple ML algorithms combined with different molecular representations (RDKit descriptors and Morgan fingerprints). The optimal model was identified and rigorously interpreted using SHapley Additive exPlanations (SHAP) to elucidate critical structure−property relationships. The model's predictive reliability was further validated against newly reported SAM molecules. Finally, we demonstrate its practical utility by designing new SAM candidates with predicted PCEs approaching 27%. This study provides an efficient computational strategy to guide the experimental development of next-generation SAM-based photovoltaic devices.

Table 1. Performance Metrics of the 12 Models

		RMSE	R^2	r	MAE
XGBoost	RDKit	1.862	0.5058	0.8161	1.528
	Morgan_256	1.904	0.4835	0.7734	1.467
	Morgan_512	1.934	0.4670	0.7752	1.416
	Morgan_1024	1.909	0.4804	0.7046	1.502
RF	RDKit	2.047	0.4031	0.6535	1.443
	Morgan_256	2.016	0.4208	0.7004	1.453
	Morgan_512	2.002	0.4290	0.6999	1.493
	Morgan_1024	1.891	0.4904	0.7179	1.378
GBDT	RDKit	1.918	0.4757	0.6997	1.199
	Morgan_256	1.900	0.4858	0.7137	1.274
	Morgan_512	1.934	0.4672	0.6941	1.363
	Morgan_1024	1.876	0.4985	0.7412	1.424

The workflow of this study is illustrated in Figure 1 and consists of the following steps. (1) Data Collection: Gathering relevant data to construct a database. (2) Feature Engineering: Selecting appropriate input features for the model. (3) Model Training: Selecting a combination of multiple machine learning algorithms and diverse input features, training various models and evaluating their performance. (4) Model Explanation: Using SHAP to interpret the model, analyzing

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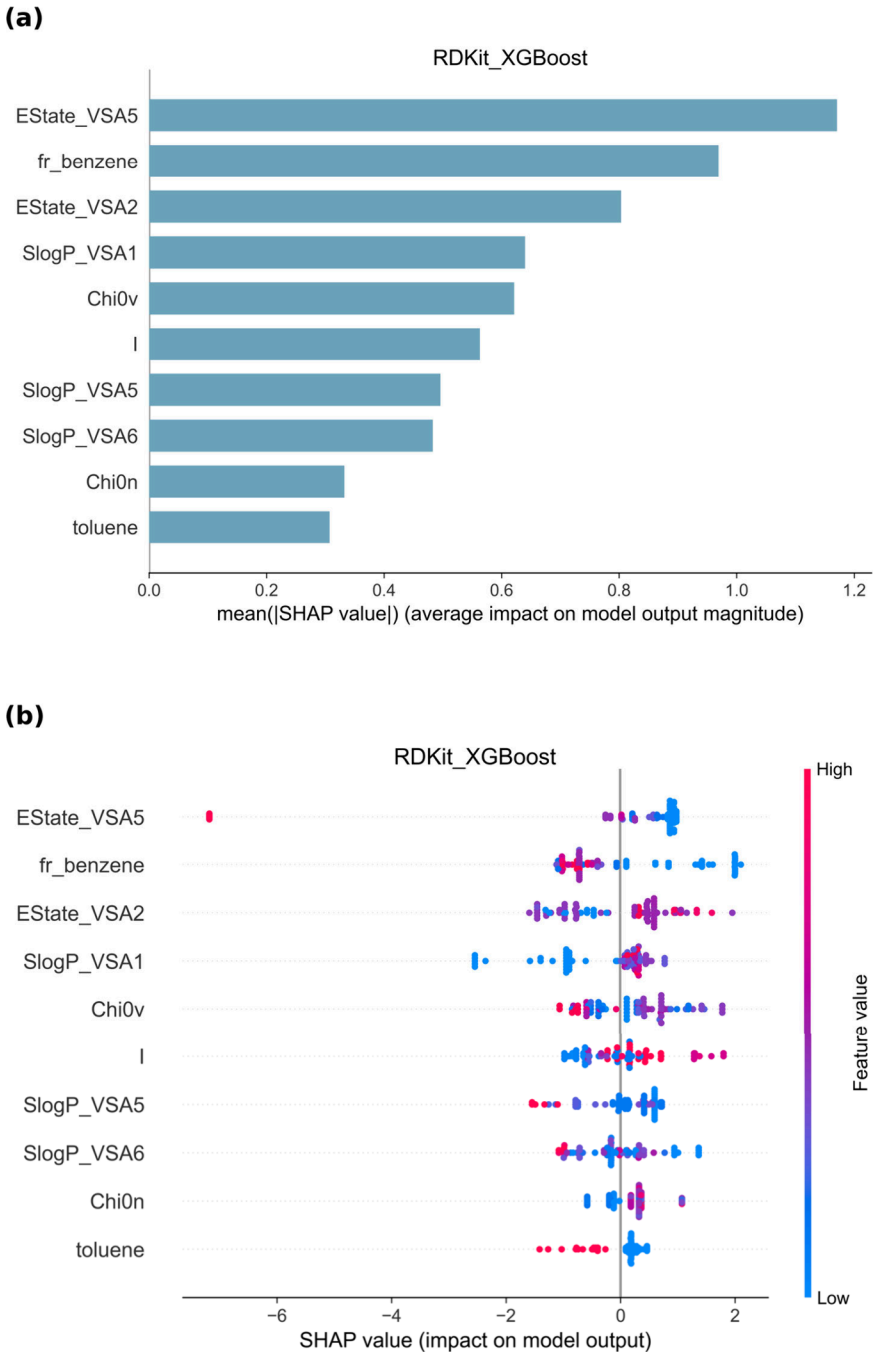


Figure 4. Importance and contribution of the top 10 features to RDKit&XGBoost model decision. (a) Summary bar plot of RDKit&XGBoost. (b) Swarm summary plot of RDKit&XGBoost.

feature importance and their relationships with the target output. (5) Model Validation: Validating the models using recently synthesized and published SAM molecules and newly designed SAM molecules to further assess their predictive performance and application potential. To investigate the relationship between the molecular characteristics of SAM materials and the PCE of inverted PSCs, we collected data exclusively from inverted PSC devices based on SAMs. We systematically reviewed research papers on SAM materials published between 2019 and 2024, extracting a total of 108 data points. For data cleaning, we excluded data points with PCE values below 10%. This threshold was chosen because such low-efficiency devices typically reflect nonoptimized fabrication conditions (e.g., incomplete layer formation, poor contact, or nonstandardized measurement protocols), rather than the inherent performance limit of the SAM material itself. Including these points would introduce noise and bias the model toward underestimating PCE, as the model would attempt to fit outliers that do not represent the true structure−performance relationship of well-functioning devices. After this exclusion, we retained 99 high-quality data points, covering more than 50 distinct SAM molecules. The excluded 9 entries are listed in Table S1 along with their PCE values. Figure S1 shows the PCE distribution of the processed data set used for our model training. The complete set of SAM molecules involved in this study is provided in Table S2.

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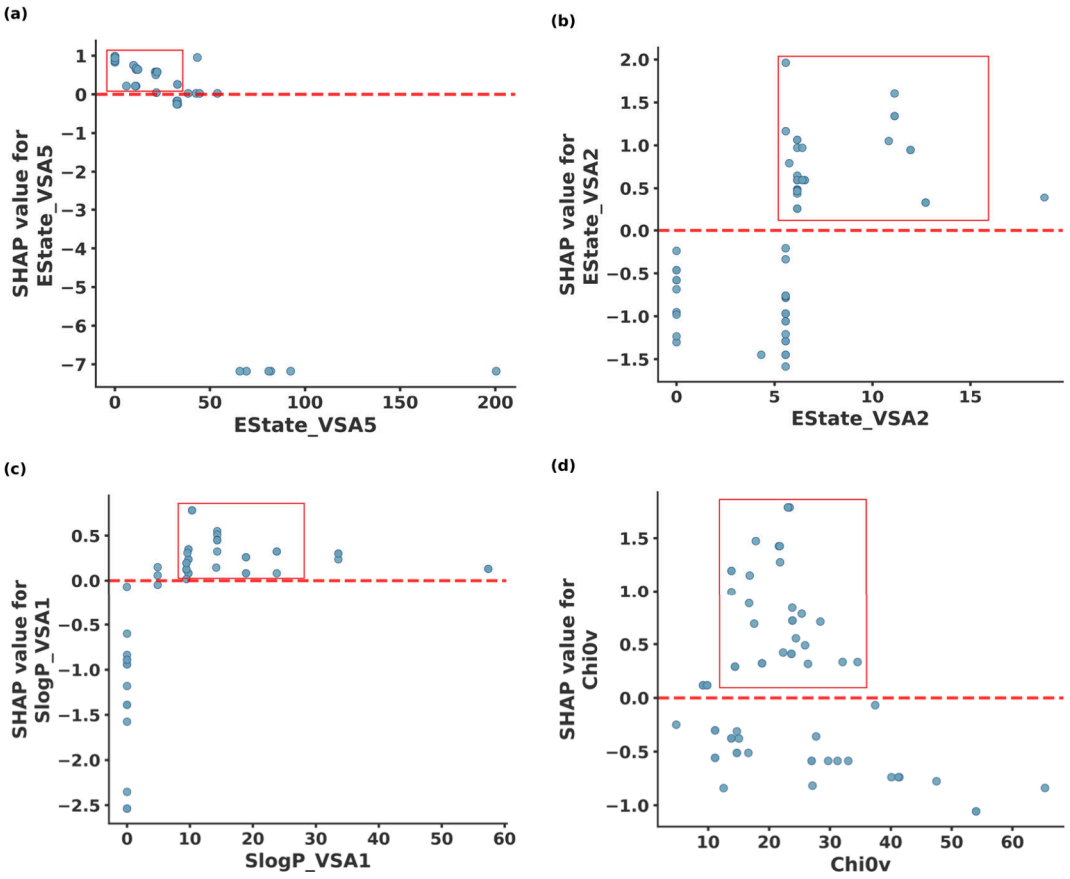


Figure 5. Effect of a single feature on the model decision. (a) The contribution of the EState_VSA5 to the model. (b) The contribution of the EState_VSA2 to the model. (c) The contribution of the SlogP_VSA1 to the model. (d) The contribution of the Chi0v to the model.

To accurately model the relationship between the molecular structure of SAM materials and PCE, appropriate input features must be selected. We obtained the SMILES (Simplified Molecular Input Line Entry System) representations of all SAM molecules and converted them into Morgan fingerprints and RDKit descriptors using Python and the web-based tool http://www.scbdd.com/chemdes/. These were used as input features for the machine learning models. Morgan fingerprints encode the presence or absence of molecular substructures in a binary format, where "1" indicates the presence of a feature and "0" its absence. As a type of circular fingerprint, Morgan fingerprints capture localized structural fragments in a flexible and detailed manner, with adjustable lengths to balance information richness and computational efficiency.[26] In this study, we tested Morgan fingerprints with hash sizes of 256, 512, and 1024 to evaluate their impact on model performance. Molecular descriptors, in contrast, represent molecular properties using real-valued numbers rather than structural encoding. They quantify physicochemical and topological characteristics (e.g., atom types, bond connectivity, etc.) to provide a numerical representation of molecular features.[27] In addition to the SAM-derived features, we incorporated 26 additional input parameters related to device fabrication and composition. These parameters can be broadly categorized into: (1) perovskite composition (e.g., types of cations and halides), (2) solvent and additive information (e.g., solvent type, concentration), (3) electron transport layer components, and (4) processing conditions (e.g., whether NiO_2 is used, whether cleaning is performed). A complete list of these 26 features, including their descriptions and categories, is provided in Table S3. Categorical variables (e.g., presence/absence of a specific component) were encoded using one-hot encoding to ensure they could be properly handled by the machine learning models. The RDKit descriptors-based final feature set for model training consisted of 222 features: 196 RDKit molecular descriptors and 26 device-specific parameters (Table S3). While this represents a high-dimensional space relative to our data set of 99 samples, we note that RDKit descriptors are curated chemical informatics features with well-defined physicochemical interpretations. Therefore, the RDKit descriptors were used directly without dimensionality reduction or explicit feature selection. To address potential overfitting concerns inherent in such a scenario, we relied on the built-in regularization capabilities of our chosen algorithms and rigorous validation protocols. By analyzing these features, we aim to elucidate their influence-both qualitatively and quantitatively on the PCE of SAM-based inverted perovskite solar cells. This approach will provide actionable insights to guide the experimental synthesis of SAM materials and the optimization of device fabrication processes.

Decision trees represent a nonparametric supervised learning approach in machine learning that solves classification and regression problems by summarizing decision rules from feature data and representing these rules in a tree-like graph structure. This algorithm can readily handle irrelevant features while demonstrating robust performance with unknown samples.[28] In this study, we employed three decision tree-based ML algorithms: Extreme Gradient Boosting (XGBoost),

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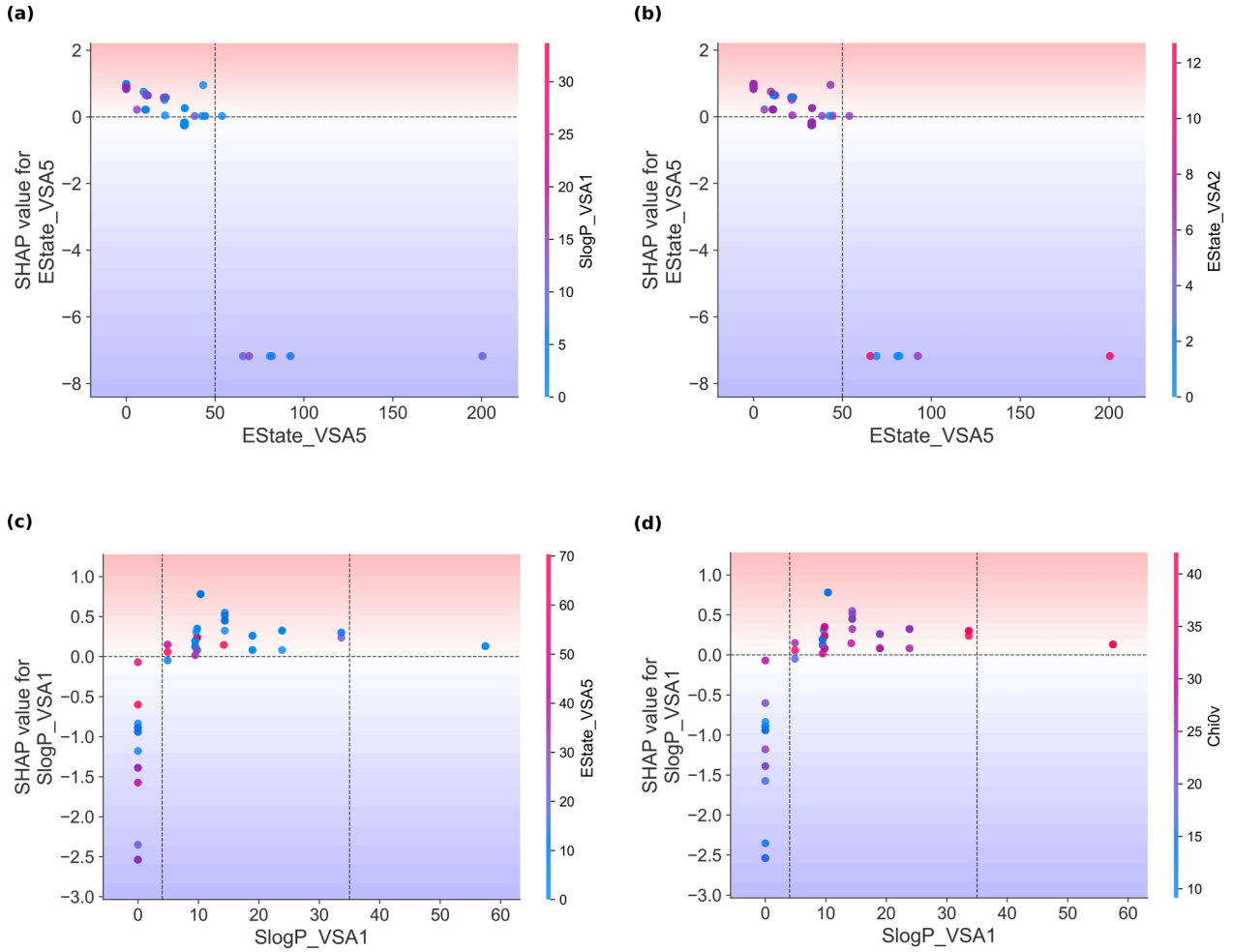


Figure 6. Effect of dual feature interactions on the model decision. (a) EState_VSA5-associated SlogP_VSA1. (b) State_VSA5-associated State_VSA2. (c) SlogP_VSA1-associated EState_VSA5. (d) SlogP_VSA1-associated Chi0v. The *x*-axis denotes the values of the primary variables, with the corresponding SHAP values plotted on the *y*-axis. The intensity of the scatter points' coloration corresponds to the magnitude of a secondary variable, as indicated by the color bar. A shifting red-to-blue background encodes the polarity of the SHAP values, where the neutral white baseline represents zero contribution (SHAP value = 0).

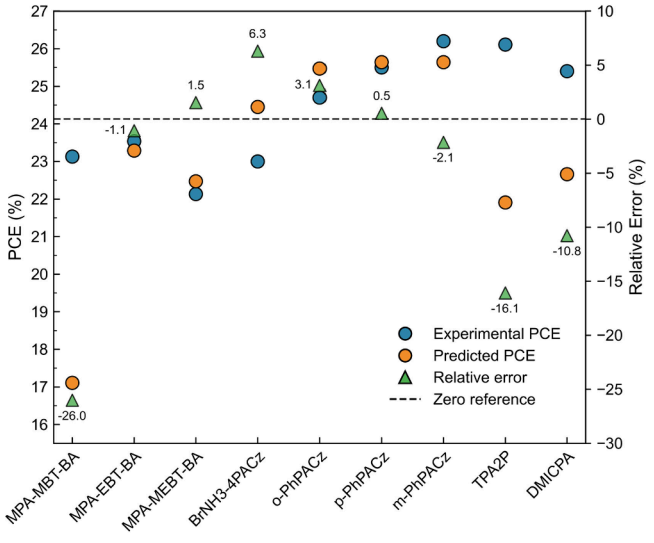


Figure 7. Model validation results of RDKit-based XGBoost model: Comparison of the predicted PCE values from the model with the experimental PCE values for the validation data set and relative errors.

Table 2. Results of the Model Validation

SAM	Experimental PCE (%)	Predicted PCE (%)	Relative Error (%)
MPA-MBT-BA	23.13[34]	17.11	26.03
MPA-EBT-BA	23.54[34]	23.29	1.06
MPA-MEBT-BA	22.13[34]	22.47	1.54
$BrNH_3$-4PACz	23[35]	24.45	6.30
o-PhPACz	24.7[36]	25.47	3.12
p-PhPACz	25.5[36]	25.64	0.55
m-PhPACz	26.2[36]	25.64	2.14
TPA2P	26.11[37]	21.91	16.09
DMICPA	25.4[37]	22.66	10.79

Random Forest (RF), and Gradient Boosting Decision Tree (GBDT).

XGBoost is an efficient ensemble learning algorithm based on gradient boosting, whose core concept involves iteratively training decision trees to optimize predictive performance. Unlike conventional gradient boosting methods, XGBoost incorporates regularization terms in the objective function to control model complexity and employs second-order Taylor expansion for more precise loss function optimization. The

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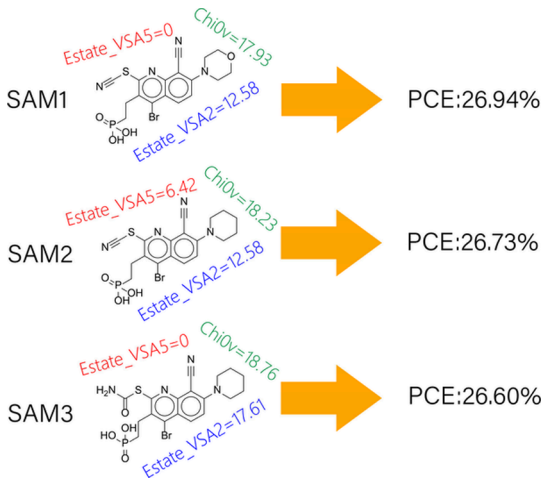


Figure 8. Three designed SAM molecules and their predicted results. The common anchoring group is a phosphonic acid. Key structural variations lie in the headgroup and the spacer/linker. SAM1 features a morpholine headgroup and a thio-cyanate side chain. SAM2 replaces the morpholine with a piperidine ring. SAM3 further modifies the headgroup to a piperidine and substitutes the thio-cyanate with a thio-amide group.

algorithm selects optimal split points through a greedy strategy, supports parallel computing, and automatically handles missing values, making it particularly effective for classification, regression, and ranking tasks.

RF is a bagging-based ensemble learning method that enhances model robustness by constructing multiple independent decision trees and combining their predictions. During training, each tree is trained on a randomly sampled subset of data (with replacement) and considers only a random selection of features at each node split to promote model diversity. The final prediction is determined through voting (for classification) or averaging (for regression), significantly reducing overfitting risks. Known for its simplicity, parallelizability, and noise resistance, RF is widely applied to high-dimensional data and nonlinear problems.

GBDT is a boosting-based ensemble algorithm that sequentially trains decision trees to correct errors from previous iterations. GBDT adopts a stepwise training strategy where each new tree fits the negative gradient (pseudo-residuals) of the current model, progressively refining predictions. The algorithm supports various loss functions and balances bias-variance trade-offs by adjusting tree depth and learning rate.

We then divided the entire data set into 80% for training and 20% for testing. Given the limited number of unique SAM molecules (over 50) and the fact that some molecules appear in multiple data points (due to variations in device fabrication or measurement conditions), we adopted a point-level splitting strategy. This means that data points from the same SAM molecule could appear in both the training and test sets, which reflects real-world scenarios where the same material may be tested under different conditions. While this approach carries a risk of data leakage, it allows us to utilize the full data set for training and evaluation. To mitigate this risk, we employed rigorous cross-validation and external validation on entirely unseen SAM molecules. Given the feature-to-sample ratio, particular attention was paid to mitigating overfitting. The tree-based models were trained with their respective regularization mechanisms enabled. For XGBoost, this includes L1 (reg_alpha) and L2 (reg_lambda) regularization terms within its objective function. Hyperparameter optimization via GridSearchCV explicitly tuned these parameters alongside others controlling tree depth (max_depth) and learning rate (learning_rate), ensuring a balance between model complexity and generalization ability. Subsequently training three ML algorithms (XGBoost, RF, and GBDT) using both RDKit descriptors and Morgan fingerprints of different lengths (256, 512, and 1024 bits) as input features, resulting in a total of 12 trained models. For each model, we optimized the hyperparameters using GridSearchCV and evaluated their performance based on four metrics: RMSE, R^2, r and MAE, to ensure the highest possible model accuracy. Meanwhile, during the model construction process, standardization was applied to the features to eliminate the influence of unit and scale differences among variables and to ensure each feature dimension is treated equally. The standardization formula is shown in eq 1, where D_{sta} is the standardized feature data, $\overline{D}$ is the mean value of the feature data, and $\sigma(D)$ is the standard deviation of the feature data. Throughout the modeling process, the machine learning models were developed using the Python programming language. The performance on a held-out test set and, more importantly, on an entirely external validation set provides empirical evidence of the model's robustness.

$$D_{sta} = \frac{D_i - \overline{D}}{\sigma(D)} \tag{1}$$

The performance of the established model was evaluated using the Root Mean Square Error (RMSE), the Coefficient of Determination (R^2), the Pearson Correlation Coefficient (r) and the Mean Absolute Error (MAE). The calculation methods for each metric are shown in eqs 2−5, where n represents the number of data entries, x_i denotes the PCE values reported in the database, $\overline{x}$ represents the average of all PCE values in the database, y_i indicates the PCE values predicted by the ML model, and $\overline{y}$ denotes the average of the PCE values predicted by the ML model. The RMSE is an important metric for evaluating the performance of regression models. It measures the standard deviation between predicted and actual values, providing a quantification of the actual prediction error. R^2 reflects the interpretability of the model. Its value ranges between 0 and 1, with values closer to 1 indicating a better fit of the model to the data. R^2 is a standardized measure of model interpretability, while RMSE quantifies the actual prediction error. These two metrics can be used together to comprehensively assess the performance of the model. The r is a statistical measure of the linear correlation between two variables. It ranges from −1 to 1, reflecting the strength and direction of the linear relationship between the two variables. MAE is a commonly used metric for evaluating regression models, which measures the average magnitude of absolute differences between predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2} \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(\overline{y} - x_i)^2} \tag{3}$$

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$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \quad (4)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \bar{y}| \quad (5)$$

After identifying the optimal model, we conducted model interpretation to examine the influence of individual features on the predictive outcomes and understand how these features affect the model's performance. Machine learning models often function as "black boxes"—while we can observe their output predictions, the underlying relationships between input features and model results remain obscure. Traditional feature importance methods can only indicate which features are more significant but fail to reveal their specific impact on model outputs. To address this limitation, we employed SHapley Additive exPlanations (SHAP), an interpretability framework rooted in game theory that provides comprehensive explanations for machine learning model outputs.[29] The core of this approach lies in Shapley values, a concept originally developed by Lloyd Shapley in 1953 to fairly distribute payoffs among participants in cooperative games.[30] SHAP quantifies each feature's contribution to model predictions by calculating its marginal impact across all possible feature combinations, ensuring equitable attribution of influence among all variables. This method offers dual capabilities: it enables both local interpretation of individual predictions and global model interpretation through aggregation of multiple Shapley values.[31]

Figure 2 presents the PCE prediction curves for all 12 models, showing that most data points closely align with the predicted vs actual fitting curves, indicating good overall performance. All optimized hyperparameters obtained from the final model are presented in Table S4.

Figure 3 displays the performance results (RMSE, MAE, R^2, and r) for each model. The heatmaps reveal that XGBoost with RDKit achieved the lowest RMSE (1.862), followed by Morrgan_1024-based GBDT (1.876). For R^2, XGBoost and GBDT significantly outperformed RF, with RDKit-based XGBoost showing the highest R^2 (0.5058). Similarly, for the r metric, RF models consistently underperformed compared to XGBoost and GBDT, with the best XGBoost model (RDKit input) achieving r of 0.8161 and the top GBDT model (Morgan_1024 input) reaching r of 0.7412. In terms of MAE, the performance rankings of the three models were distinct. The GBDT model achieved the lowest MAE values across most feature sets, followed by the RF model, while the XGBoost model consistently exhibited the highest MAE. Overall, the RDKit-based XGBoost model demonstrated superior performance with RMSE of 1.862, R^2 of 0.5058, r of 0.8161 and MAE of 1.528, while RF models performed significantly worse than both XGBoost and GBDT. The complete performance metrics for all models are summarized in Table 1. Meanwhile, Figure S5 further presents the performance metrics of the 12 models after 10-fold cross-validation. Collectively, the RDKit-based XGBoost model performed the best.

Figure 4 presents the SHAP interpretation results for the optimal model in this study-the RDKit-based XGBoost model (Figures S2−S12 display the SHAP interpretation results for the other models). Figure 4a shows the importance ranking of each feature in the model, where the vertical axis represents each feature (ranked from top to bottom, with higher positions indicating greater importance to the model results). The horizontal axis shows the mean absolute value of Shapley values for each feature, representing their average influence on the model output. From the ranking, we can see that the top 5 features are all RDKit descriptors, namely EState_VSA5, fr_benzene, EState_VSA2, SlogP_VSA1 and Chi0v, which have a significant impact on the model output, and the combined influence of these five features far exceeds that of other features. This indicates that changes in the model's output are largely affected by these few features. Furthermore, among the top 10 important features, nearly all are RDKit descriptors, suggesting that molecular descriptors play a dominant role in determining the photovoltaic performance prediction in our model. Figure 4b further explains how each feature influences the model output by analyzing their positive or negative contributions to the predictions using the SHAP additive explanation algorithm. In this result plot, each row represents a feature, and each dot represents a data point. The features are again ranked in descending order of their importance to the model output. The horizontal axis displays both positive and negative SHAP values, with red portions indicating high feature values and blue portions indicating low feature values. The results show that for the top-ranked feature EState_VSA5, high values (red) cluster in the negative SHAP region, indicating a detrimental effect on PCE, while low values (blue) exert a positive effect. For EState_VSA2, the SHAP result shows an opposite trend. It can be observed from the SHAP analysis that when its feature value is low, it exerts a negative contribution to the model's predicted PCE outcome, whereas a higher feature value results in a positive contribution. The relationships for Chi0v and SlogP_VSA1 are more complex. For Chi0v, an intermediate range of values yields a positive contribution to the model's output, while values that are either too low or too high exhibit a negative impact. Similarly, SlogP_VSA1, shows low feature values contributing negatively and high values contributing positively in SHAP analysis, suggesting that an appropriate level of hydrophobicity is beneficial for enhancing PCE.

Expanding the physical interpretation, these SHAP-derived trends can be linked to plausible SAM-perovskite interface mechanisms. EState_VSA5, quantifying specific atomic electrotopological contributions to molecular surface area,[32] correlates negatively with PCE when high. This suggests that excessive localized polarity may disrupt optimal interfacial dipole formation and energy level alignment, which iscritical for efficient charge extraction. SlogP_VSA1, a descriptor of hydrophobic surface area,[33] shows a positive correlation beyond a threshold. Adequate hydrophobicity likely enhances interfacial moisture resistance and promotes denser SAM packing, both vital for device stability and uniform charge transport. Chi0v, a valence connectivity index related to molecular size and complexity,[32] exhibits an optimal intermediate range for PCE. This implies a balance is needed: sufficient molecular size for effective interface modification, yet avoiding excessive bulk that could hinder monolayer order or introduce defects. While these descriptor-property links offer valuable design insights, they represent statistical correlations whose precise atomic-scale mechanisms warrant further experimental validation.

Figure 5 further demonstrates the influence of individual features on the model's output. As shown in Figure 5a, for the feature EState_VSA5, the model considers it to have a positive

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contribution to predicting PCE when its value falls within the approximate range of 0 to 50. Furthermore, when its value is 0, the positive contribution to the model reaches its maximum. When the value exceeds 50, the feature's contribution shifts from positive to negative. Regarding the feature EState_VSA2, the threshold is approximately 6. Values below 6 contribute negatively to the model output, whereas values above 6 exert a positive influence. Similarly, for the feature SlogP_VSA1, the SHAP dependence plot reveals a threshold around 5. Values below this threshold exert a negative contribution to the model output, while values above it contribute positively. For the feature Chi0v, its contribution to the model is positive when its value lies within the 15−25 range. This positive impact is maximized at values around 20. Conversely, values outside this range have a negative impact on the model's performance. Through this SHAP interpretation of the model results, we have further demonstrated the degree of influence each feature has on the PCE output of our optimal model and how they affect the model's predictions.

The interaction plot can reveal the marginal effects of one or two features on a machine learning model's prediction, illustrating both the relationships between features and the target variable, and the interactions between features themselves. To further elucidate the pairwise interactions between different descriptor features and their impact on the model output, we analyzed the SHAP interaction values for EState_VSA5, EState_VSA2, SlogP_VSA1, and Chi0v, as shown in Figure 6. Figures 6a and 6b present the interaction effects on the model output when EState_VSA5 is considered as the main feature, paired with SlogP_VSA1 and EState_VSA2, respectively. From Figure 6a, it can be observed that a positive contribution to the PCE is achieved only when both EState_VSA5 and SlogP_VSA1 exhibit low values. When these two features take on medium or high values, their joint effect on the model output becomes negative. Conversely, Figures 6c and 6d display the interaction effects with SlogP_VSA1 as the main feature, paired with EState_VSA5 and Chi0v, respectively. From the results in Figure 6c, it can be seen that when SlogP_VSA1 takes a small value, the feature value of EState_VSA5 also needs to be small to have a positive contribution to the model. In contrast, the interaction between Chi0v and SlogP_VSA1 shows the opposite trend: as shown in Figure 6d, when SlogP_VSA1 takes a small value, the feature Chi0v needs to take a larger value to have a positive contribution to the model.

After completing the training of all models and identifying the best-performing model, the RDKit-based XGBoost model, we further validated its reliability and explored its application in screening and designing high-PCE SAM materials. We collected several newly synthesized SAM molecules reported in 2025 literature and input them into our optimal model for PCE prediction. The results are shown in Figure 7. We selected nine SAM molecules with reported PCEs between 22% and 27% for testing. Figure S13 shows the molecular structures of these SAM molecules. As shown in Figure 7, the PCE values predicted by the model for most molecules align closely with their actual reported PCEs. Through calculation, the relative errors are largely within 10%, with a minimum error of only 0.55%. Table 2 summarizes the experimental PCEs, model-predicted PCEs, and relative errors for these molecules. These results further demonstrate the reliability of our trained model in predicting PCE for SAM-based inverted perovskite solar cells. Moreover, this work provides valuable guidance and strategies for machine learning-assisted screening and design of high-PCE SAM materials.

Then, with the guidance of our model, three new SAM molecular structures were designed and designated as SAM1, SAM2 and SAM3, respectively. Their molecular structures are shown in Figure 8. Based on our previous SHAP analysis, the most important features influencing the model output are descriptors such as EState_VSA5. The EState_VSA5 values for the three newly designed SAM molecules are 0, 6.42 and 0, which not only falls within the optimal range of 0−50 as identified by the SHAP analysis but also corresponds to the condition that maximizes the feature's positive contribution to the predicted PCE; their EState_VSA2 values are 12.58, 12.58 and 17.61; Chi0v values are 17.93, 18.23 and 18.76, respectively. It can be observed that the values of these key descriptors for the SAM molecules all lie within the expected ranges indicated by the SHAP analysis. Subsequently, based on the SHAP results, appropriate values were also assigned to other features such as perovskite composition to achieve a predicted high PCE outcome. The predicted PCE values for these three SAM molecules are 26.94%, 26.73%, and 26.60%, respectively, suggesting their promising theoretical performance. These results demonstrate that our model can effectively assist in the design and screening of high-performance SAM candidates for perovskite solar cells, thereby potentially accelerating the development of SAM-based photovoltaic devices. It should be noted, however, that these predictions await experimental validation.

In this study, we compiled a data set of 99 self-assembled monolayer materials from published literature on inverted perovskite solar cells. We systematically constructed molecular representations using RDKit descriptors and three variants of Morgan fingerprints, and trained a total of 12 machine learning models by combining these features with three distinct algorithms to predict the PCE of SAM-based inverted perovskite solar cells. Our results demonstrate that the XGBoost model utilizing RDKit descriptors achieved optimal performance, with r of 0.8161, RMSE of 1.862, R^2 of 0.5058 and MAE of 1.528. The SHAP analysis revealed that the top five features most significantly influencing the model's output results were all RDKit descriptors generated from SAM molecules, specifically EState_VSA5, fr_benzene, EState_VSA2, SlogP_VSA1, and Chi0v. These critical descriptors provide valuable insights for rational SAM design and screening. External validation using newly synthesized SAM materials confirmed the model's reliability. The relative errors between the predicted and experimental PCE values were mostly within 10%, with a minimum error of only 0.55%. Moreover, three new SAM molecules designed under the guidance of our model with the highest predicted PCE approaching 27%, demonstrating its significant application potential. This work establishes an efficient machine learning framework for accelerated self-assembled monolayer materials discovery and offers valuable guidance for developing high-performance organic photovoltaic materials and devices. It is worth noting that the current model is built upon a data set of 99 SAM-based PSC devices. While this data set reflects some high-quality experimental data currently reported in the field, the performance and generalizability of machine learning models can typically be enhanced by larger and more structurally diverse data sets. Future efforts to expand this database, for instance, by including newly published SAM structures and additional experimental parameters, will be

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valuable for further improving predictive accuracy and exploring a broader region of the chemical and processing space.

■ ASSOCIATED CONTENT

Data Availability Statement

The data and code can be downloaded at a GitHub repository (https://github.com/Haifeng-Li-ML/SAM-ML.git).

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.6c00119.

Nine low-PCE data points excluded from model training, the data distribution of the processed data set used for model training, molecular structures and SMILES strings of SAM molecules, 26 device fabrication and composition parameters used as input features, hyperparameter tuning results for all 12 ML models, 10-fold cross-validation performance metrics, SHAP analysis results for 11 additional models, molecular structures of nine validation SAMs, and complete reference list for SAM data set (PDF)

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Notes

The authors declare no competing financial interest.

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